Ishitha Padmaraju

Data Scientist

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Profile

Data Scientist with 3+ years of experience building and deploying machine learning solutions across industrial, automotive and insurance domains. Implemented real-time email classification and anomaly detection with edge deployment and cloud scalability.

Eager to bring my experience in deep learning and applied AI research to contribute to high impact analytics transformations across industries. Open to relocate across Germany.

Primary stack: Python, PyTorch, SQL, Docker, AWS, FastAPI, Git, Airflow

Experience

10/2025 – 03/2026 **Data Scientist - ML**

Check24 GmbH *Frankfurt, Germany*

- Owned the end-to-end development of ML models for automated classification of ~15,000 weekly emails enabling scalable automation.
- Built a production-grade ML pipeline mapping unstructured document content to product and intent (e.g. damage), improving routing speed.
- Increased precision for 5 document types by combining ML models with rule-based validation logic, supporting compliance-relevant workflows.
- **Skills**: Python, SQL, AWS, Git, Airflow, Prompt Engineering

02/2024 – 01/2025 **Machine Learning Researcher**

Schaeffler Hub for Advanced Research at KIT *Karlsruhe, Germany*

- Accelerated real-time inference for perception tasks to 187 FPS with GPU optimization, cutting latency by 99.98%.
- Led end-to-end development of a sensor fusion framework using ROS 2 synchronizing sensor data for real-time prediction achieving 91% accuracy.
- Containerized best preforming model with Docker image in Harbor registry for scalable inference on an embedded application.
- Reduced memory consumption by 50% through precision tuning, layer-wise profiling, and targeted quantization of deep learning models.
- **Skills**: Python, OpenCV, PyTorch, CUDA, Shell, GitLab CI/CD, Cursor

08/2023 – 01/2024 **R&D E-mobility Systems Intern**

Schaeffler Technologies AG & Co. KG *Herzogenaurach, Germany*

- Developed an AI model as a proof of concept for anomaly detection for test benches by time series analysis which increased usage efficiency by 24%.
- **Skills**: Python, Scikit-learn, Pandas, PowerBI, GitLab CI/CD

10/2021 – 09/2022 **Machine Learning Engineer**

Stellantis *Hyderabad, India*

- Implemented a CAN based fault detection system for 500+ vehicles with 95% prediction accuracy.
- Queried vehicle telemetry logs to extract relevant signal patterns for fault prediction and feature engineering.
- Built CI/CD pipelines for scalable cloud deployment in Agile workflows.
- **Skills**: Python, SQL, Docker, FastAPI, AWS Lambda, Sagemaker, VS Code

Education

10/2022 – 09/2025 **Master in E-Mobility – Artificial Intelligence and Autonomous Driving**

Friedrich-Alexander University Erlangen-Nuremberg *Germany*

Key Courses: Data Science, Machine Learning, Deep Learning GPA: 1.6

Thesis: DL-based Road Condition Prediction from Distributed Sensor Data

07/2017 – 07/2021 **Bachelor in Mechanical Engineering**

Jawaharlal Nehru Technological University Anantapur *India*

Key Courses: Robotics & Automation, Operations Research GPA: 1.7

Top 5 % in the batch

Projects

2025 **Lead Scoring & Customer Insights Engine**

- Developed a RAG pipeline using an LLM (GPT-3.5-turbo) and vector search (FAISS) to analyze 1000+ documents for lead scoring and insights.
- **Skills**: Python, RAG, Vector DBs, LangChain, Streamlit, AWS Lambda

2024 **AI Design Generator (Wunderhub)**

- Spearheaded a PoC, trained a ViT model using PyTorch & Hugging Face, with 85% semantic alignment in graphics generated from text prompts.
- **Skills**: PoC development, PyTorch, Hugging Face, Flask APIs, NLP

Languages

English (bi-lingual), German (B2), Telugu (bi-lingual), Hindi (B2)

Hobbies

Learning	Participation in Tech Meetups and Workshops to learn new technologies
Lifestyle	Culinary Experiments and Yoga/Meditation for mental balance

Date: 04.03.2026 Signature: Oshith Padmaraju